

PROFESSIONAL SUMMARY

Computer Engineering student with hands-on experience in embedded systems, IoT telemetry, electrical prototyping, full-stack digital systems, and hardware-software troubleshooting. Experienced in building operational dashboards, enterprise workflow systems, digital twin prototypes, and field-oriented automation projects for reliability, monitoring, and process digitalization.

Technical Skills

- o **Programming Languages:** Python, C/C++, JavaScript/TypeScript, PHP, Dart.
- o **Embedded & Hardware:** ESP32, STM32, ATmega16, Raspberry Pi 4B, PCB Design & Soldering, Sensors, Motor Control, Sensor Integration, Power Electronics (24V BMS, DC-DC, Relay)
- o **IoT & Systems:** MQTT, GSM/nRF Telemetry, Digital Twin (PoC), Real-time Monitoring Dashboard, Real-time control dashboard, Edge Computing, Sensor Integration.
- o **AI / ML:** Random Forest, Reinforcement Learning, Sensor-based Decision Support.
- o **Frontend & Mobile:** Flutter, React.js, Leaflet.js (GIS), Tailwind, Figma (UI/UX)
- o **Backend & API:** Node.js, Laravel, PERN Stack, REST API, FastAPI, Cloudflare R2, LLM API
- o **DevOps & Tools:** Docker, Git/GitHub, PLC Programming (basics), VS Code, AntiGravity
- o **Domain Knowledge:** ERP/HRIS Module Development, Workflow & Process Digitalization, Business Process Modeling, Technical Documentation & SOP.

WORK EXPERIENCES

PT. Saraswanti Utama Group

Surabaya, ID

Software Engineer Intern

Jan 2026 – Mar 2026

- o Developed Flutter frontend modules for an enterprise HRIS used by hundreds of employees, including Auth, Leave/Permission Management, Asset Reservation, Approval Workflow, and E-Catalog.
- o Implemented complex routing and IndexedStack-based state preservation to maintain user sessions and improve navigation across multiple HRIS modules.
- o Contributed to the Transport Management System as part of the company's internal SCM ERP, supporting logistics tracking and operational workflow digitalization.
- o Supported internal application troubleshooting by diagnosing UI behavior, module flow issues, user feedback, and operational bugs during HRIS/TMS development and testing.
- o Prepared technical notes and workflow documentation to support maintainability, user understanding, and handover of internal enterprise system modules.

PV-Cloost (Energy Startup)

Surabaya, ID

Embedded Software Engineer

April 2026 – Present

- o Designed and tested embedded and electrical components for an industrial-scale remote-controlled solar panel cleaning robot, focusing on reliability, durability, and outdoor deployment readiness.
- o Developed robot control firmware based on operational requirements, including motor control, remote-command handling, safety behavior, and system reliability.
- o Supported electrical prototyping and validation, including ESP32-based PCB testing, 8S 24V battery system, 24V high-torque encoder motors, DC-DC step-down modules, fuses, switches, cabling, and power distribution.
- o Performed hardware-software troubleshooting for robot subsystems, including motor response, remote-control behavior, power stability, wiring issues, and component-level failures during testing.
- o Documented testing results and technical findings to support iteration, maintainability, and real-world deployment preparation.

M-IOT Lab

Surabaya, ID

Computer Network Practicum Assistant

May 2026 – June 2026

- o Assisted computer network practice sessions for 28 teams / 80+ students, covering hands-on configuration and troubleshooting of MikroTik-based network infrastructure.
- o Developed and revised practicum modules for Wireless Networking, Firewall & NAT, VPN, and QoS using MikroTik core routers, core switches, access points, and related network devices.
- o Supported students in diagnosing connectivity and configuring, subnetting IPV4, IPV6, routing, firewall, NAT, wireless, VPN, and QoS configuration issues through structured lab guidance.

Basic Programming Practicum Assistant

Sep 2025 – Dec 2025

- o Facilitated weekly lab sessions for 12 teams / 60+ students over a 5-week period, guiding programming fundamentals, software development, and real-time debugging.
- o Assisted final-project development for simple game applications as a comprehensive test of programming logic.

- Built a digital-twin prototype for a hydropower plant (PLTA) at the Dieng site: defined end-to-end concept & architecture from developer workflow to user workflow (prototype/PoC level).
- Feature engineered an RL learning pipeline and achieved ~80% performance (internal evaluation).
- Developed a web UI for model outputs (FastAPI + JavaScript/TypeScript) and designed the UI/UX in Figma.
- Implemented live-selection and scenario-prediction visualization using Leaflet.

PERSONAL PROJECTS

- **Rice Milling Management System (On Going)** – Full-stack Developer & System Analyst **2026**
 - Built business process flows based on direct field interviews and surveys, then translated real UMKM operations into standardized yet adoptable system workflows verified by field users.
 - Developed a custom ERP system using PERN Stack and Cloudflare R2, covering procurement, production, inventory, sales, and finance workflows for rice milling operations.
 - Built an AI Analyst backend feature using Claude LLM API and automated formatted PDF generation for structured financial reporting and business process digitalization.
- **Intelligent AWD-Based Precision Irrigation DSS for Rice Fields** – System Designer & Developer **2026**
 - Designed an Indonesia-focused AWD irrigation DSS inspired by RYNAN AWD+, adapting smart water-level monitoring and irrigation recommendation concepts for local rice-field conditions.
 - Developed a cost-efficient sensing node concept using STM32, BMP/DHT/HC-SR04 sensors, current/voltage sensing, battery supply, mini solar panel, nRF local RF, and GSM cloud telemetry.
 - Designed the software architecture using PERN, FastAPI REST API, WebODM/GIS processing, Leaflet geo visual, mastering database, Cloudflare R2, and BMKG weather API to support weather-aware recommendations and interactive geospatial visualization.
 - Simulate real condition of watershed in ricefield using specific algorithm to do autorouting the best-case scenario waterflow to generate best recommendation output command as decision support system.
- **ATmega16 Minimum System & High-Speed Downloader** – PCB Design & Manual Soldering **2026**
 - Designed and assembled an ATmega16 minimum system and high-speed downloader on perfboard, focusing on compact layout and minimal jumper usage.
 - Practiced low-level embedded workflow, including manual driver configuration with Zadig and C code compilation using WinAVR.
 - Rebuilt and troubleshot corroded board connections to achieve a compact 1.5 MHz downloader, strengthening trace-level hardware understanding and soldering precision.
- **AIoT Smart Irrigation with Soil Monitoring & Controlled Irrigation (PoC)** – System Engineer **2025**
 - Built a soil-condition monitoring and controlled irrigation prototype using 3 sensors with ~2s local dashboard refresh.
 - Integrated ESP32 and Raspberry Pi 4B using MQTT over WiFi, enabling lightweight telemetry exchange between the microcontroller layer and edge-processing layer.
 - Trained and deployed a Random Forest model on Raspberry Pi with ~80% accuracy for sensor-based decision support.
 - Implemented a monitoring dashboard using FastAPI + JavaScript/TypeScript with lightweight persistence using SQLite.
- **Rice Price/Quality Analytics (B2B)** — Full-stack Developer & System Analyst **2025**
 - Scraped multi-source data using Python. designed the end-to-end data flow (ingestion → normalization → reporting).
 - Displayed normalized tables (region, Rp/kg, quality, etc.) and Top 10 cheapest regions per quality ranking. Generated Word Cloud (SVG) + Sentiment Analysis computed via Google Gemini LLM then exported results (CSV/JSON & SVG) and persisted browser snapshots.
 - Tech Stack: Vite + React + TypeScript (PapaParse) , Node.js (ESM) + Express + Gemini Responses API
- **Jump Coach AI – AI-based Vertical jump Assistant** – Mobile Sensor Tester & UI/UX Designer **2025**
 - Designed a mobile application concept to estimate jump height using smartphone sensors, including gyroscope, barometer, and GPS, as an alternative to dedicated jump measurement tools.
 - Created 3 core modules with 8 total UI pages and developed user flow diagrams to define the end-to-end interaction flow.
 - Conducted black-box sensor testing by comparing gyroscope and barometer-based measurements, finding that barometer data produced better estimation accuracy, although availability depends on specific phone models.
- **ESP32 Autonomous Hybrid Vehicle “Red Spider”** – Systems Engineer **2025**
 - Engineered a dual-mode robotic vehicle using ESP32, ultrasonic sensors, and C++ that can switch between autonomous navigation and Flutter-based Bluetooth emergency control.
 - Developed a “Smart Reverse” obstacle evasion logic using 5 side sensor comparison to detect open paths and steer

the vehicle toward safer zones.

- Implemented a 5-sensor ultrasonic array to provide wide frontal coverage and blind-spot monitoring for collision prevention.

- **Smart Temp & Air Circulation Control System (Heater Chamber) – IoT Engineer** **2025**

- Developed a precision thermal regulation system for an incubation chamber, maintaining a stable 27.0°C (±0.5°C) using a hysteresis control algorithm.

- Integrated a PTC Heater and Exhaust Fan driven by a 2-channel relay module to automatically balance heat generation and dissipation, preventing overshoot.

- Designed a dual-interface dashboard allowing seamless switching between autonomous sensor-based control and full manual overrides for maintenance.

- **Automatic Gate (PoC) – IoT Engineer** **2025**

- Built a mini automatic gate prototype using an ultrasonic sensor to detect object movement and direction, with a servo motor simulating gate actuation.

- Developed a web-based monitoring interface with manual control capability for emergency operation.

- Stored detected object data as event records and logs for monitoring and traceability.

ORGANIZATIONAL EXPERIENCE

Mage 11

Surabaya, ID

Lead Web Developers

Jan 2025 – Nov 2025

- Spearheaded the end-to-end technical lifecycle for a new event platform with no legacy code, implementing a Hybrid Project Management approach (Waterfall for structural planning and Agile for execution) to ensure a formal yet adaptive development cycle.
- Orchestrated team management and human capital development for a cross-functional team of 11, where 90% of members were non-developers; facilitated technical teaching sessions and team-bonding initiatives to bridge the skill gap and reach project targets.

UKM Robotika ITS

Surabaya, ID

Project Officer

June 2025 – Sept 2025

- Led a 15-person team through a 3-month execution cycle, restructuring the organization into four specialized divisions (Event, Media Marketing, Materials, and Technical) to maximize operational efficiency.
- Pioneered the transition to online activities, overcoming previous offline limitations by delivering a flagship webinar for 650 participants (216% of the initial target) with a 92/100 satisfaction rate.

BEM Fakultas Teknologi Elektro dan Informatika Cerdas

Surabaya, ID

Pemandu FTEIC

Oct 2024 – Dec 2025

- Conducted 5+ leadership sessions with 30+ participants each, emphasizing interactive learning and strategic thinking.
- Designed and facilitated “discussion-like presentaton” sessions to strengthen leadership and teamwork capacity.

EDUCATION

SEPULUH NOPEMBER INSTITUTE OF TECHNOLOGY

Surabaya, ID

Bachelor of Engineering in Computer Engineering (GPA: 3.44/4.00)

August 2023 – Present

SMA PU ALBAYAN

Sukabumi, ID

Science Program (Avg. Score: 92/100)

2020 – 2023

CERTIFICATION & TRAINING

- [KodeKloud] **Cloud** - Docker Training Course May 2026
- [Udemy] **PLC Fundamental** - PLC Programming from scratch by paul lynn May 2026
- [Talentiv] **SOP** - SOP & WI Professional Training Jan 2026
- [Binus Center] **Finance** - Financial Reporting with Microsoft Excel Oct 2025
- [WebHooz] **Web Dev** - Laravel, Node.js, and React.js Training Aug 2024
- [Course-Net] **Networking** - Cisco Certified Network Associate Program July 2023
- [UPBG ITS] **Language** - TEFL ITS 547 July 2023
- [IELTS] **Language** - Band 6.0 Dec 2022

SCHOLARSHIPS & AWARDS

- HETI Student Youngpreneurship (HETI SY) Grantee 2026
- Program Pembinaan Mahasiswa Wirausaha (P2MW) Grantee 2026
- KaCer ScholarShip Awardee 2025
- Bakti BCA Scholarship 2024
- ELITS Scholarship Program 2024